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Object Placement Localization in Street Scenes: A Candidate Ranking Framework with Support-Surface Constraints

PART 1

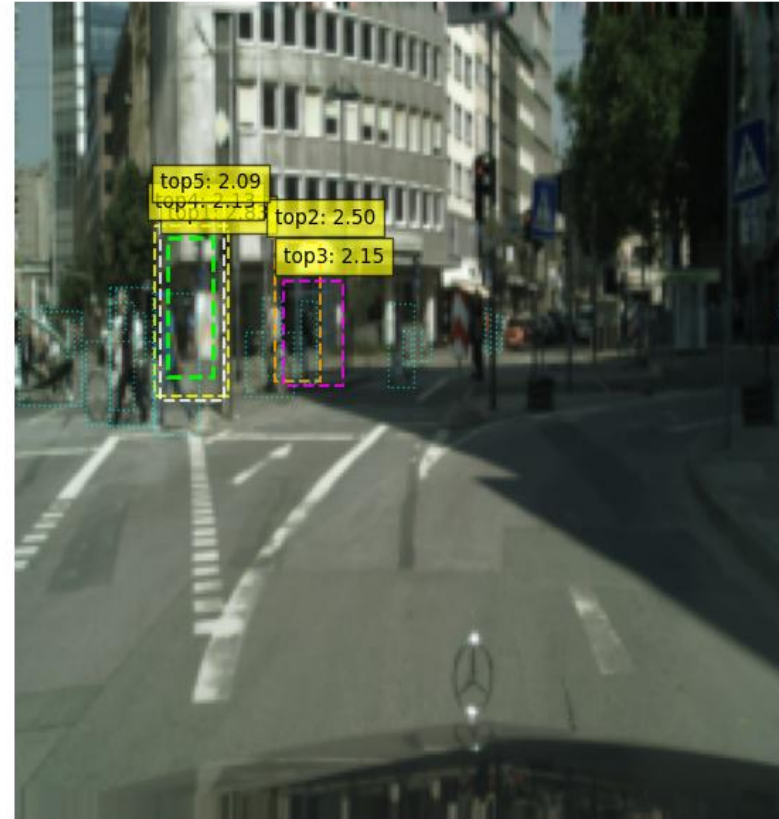
Background and Motivation

What is Object Placement?

place a person



place a person | top5



- Given a street-scene image and a text prompt, predict a bounding box for placing a new object
- NOT object detection — predicting where a new object could realistically appear
- Applications: data augmentation, autonomous driving, image editing

Key Challenges

Multiple Valid Answers

A person could stand on the left or right sidewalk — both are correct.
Direct regression averages all valid locations, producing invalid results

Semantic Constraints

Pedestrians belong on sidewalks, not on rooftops. Cars belong on roads, not on buildings

Geometric Consistency

Bounding boxes must match category-specific shapes.
Scale must be consistent with scene depth/perspective

Related Work: BOOTPLACE (Zhou et al., 2025)

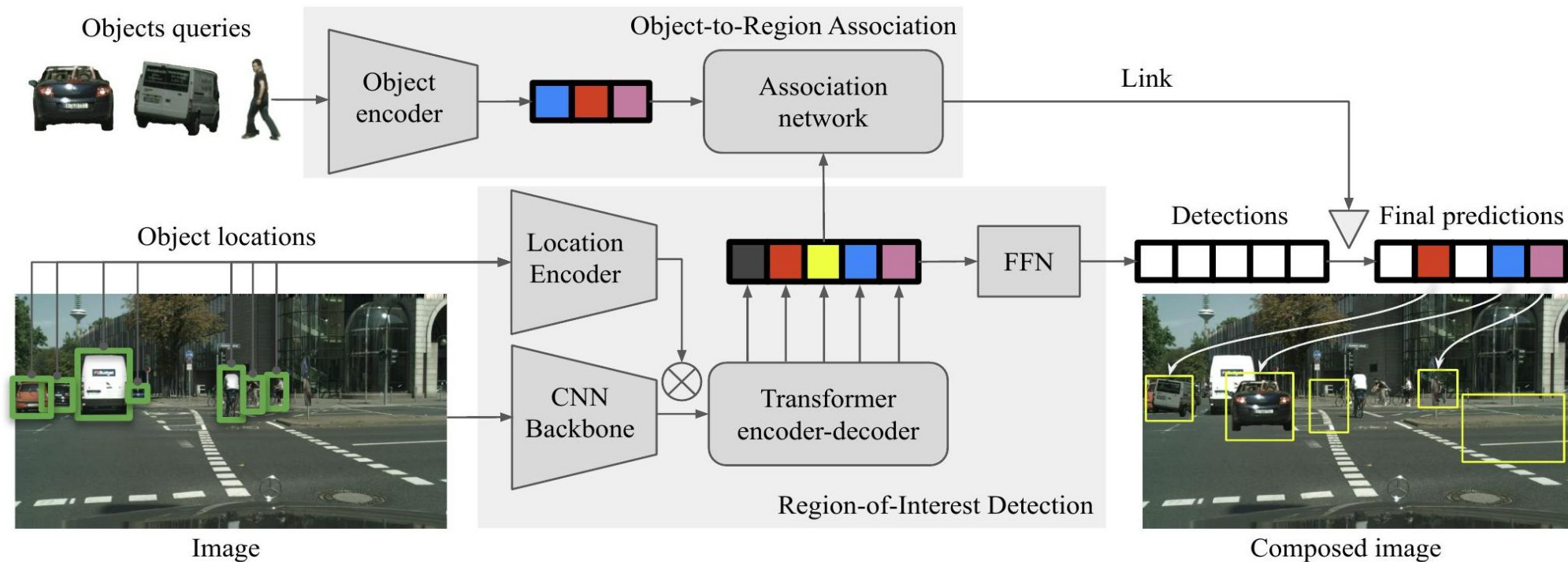


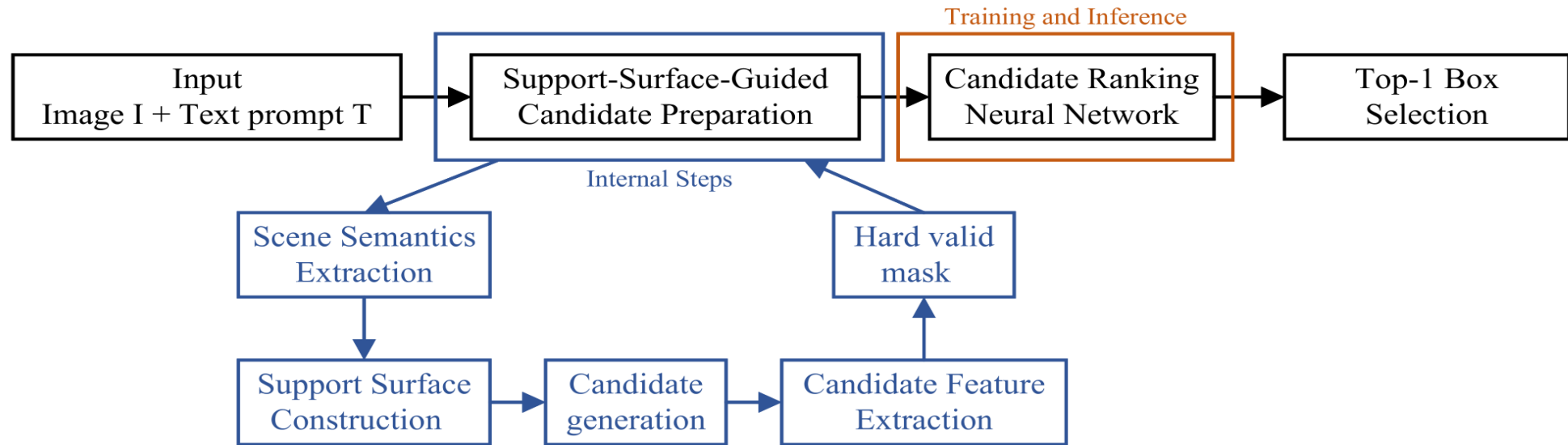
Figure 2. **Network inference.** Given a target image, several object queries (e.g., two cars and a pedestrian) and scene object locations, BOOTPLACE detects a set of candidate region of interest and associates each object with the best-fitting region, which are used to produce the composite image. $\otimes$ is feature concatenation and ∇ is region-wise product.

Formulates placement as detection on inpainted backgrounds

Bootstrapped training: randomly remove object subsets for data augmentation

Limitation: no explicit semantic constraint on candidate regions

Our Approach — Pipeline Overview



Generate candidate boxes → Score with semantic-aware features → Select top placement

Input: background image + text prompt → Output: predicted bounding box

PART 2

Methodology

Candidate Preparation

We generate **candidate boxes** guided by the target object and **support surface**, and then introduce **hard constraints** to improve qualitative plausibility.

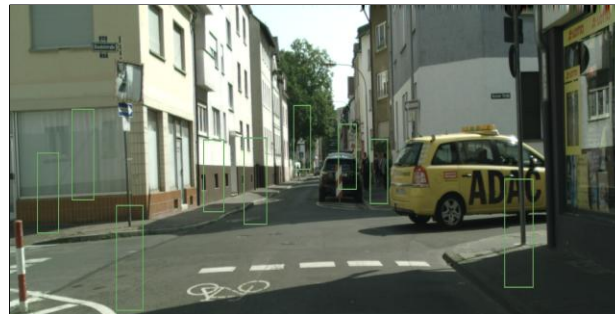
Support Surface

Identify support regions for placing the target object on scene semantics.



Candidate Generation

Generate candidate boxes around the support surface, with a small number of global fallback candidates



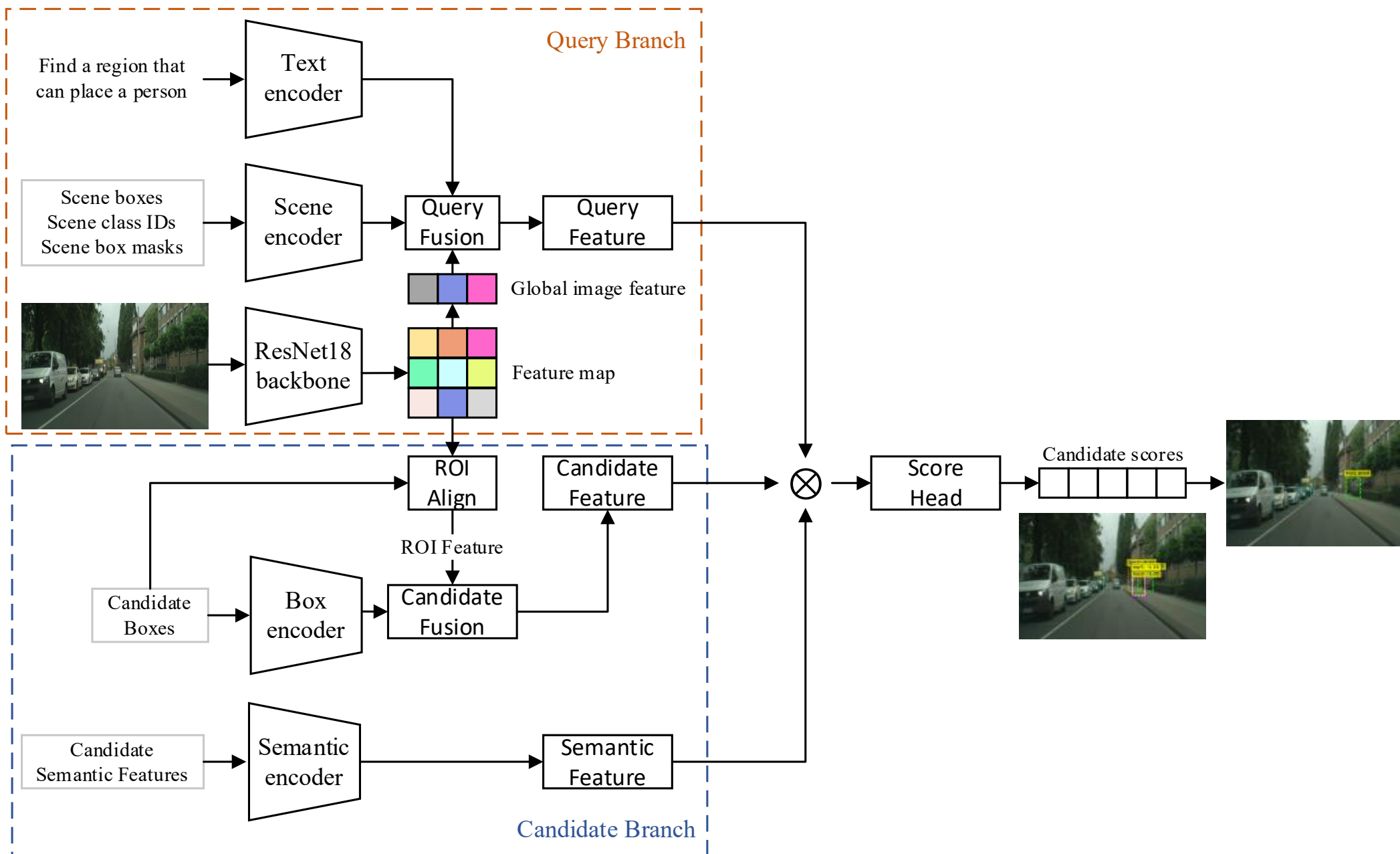
Hard Constraint

Check plausibility of geometry and class-specific semantic

- Bottom-center location
- Bottom-strip overlap



Ranking Neural Network



PART 3

Experiments, Evaluation and Conclusion

Ablation Study Design

Candidate Model

(Baseline)

Random candidate sampling,
scoring network ranks all 256
boxes

No semantic guidance

+Support
Surface

SupportSurface Model

95% candidates from support
surface, category-specific
shape priors

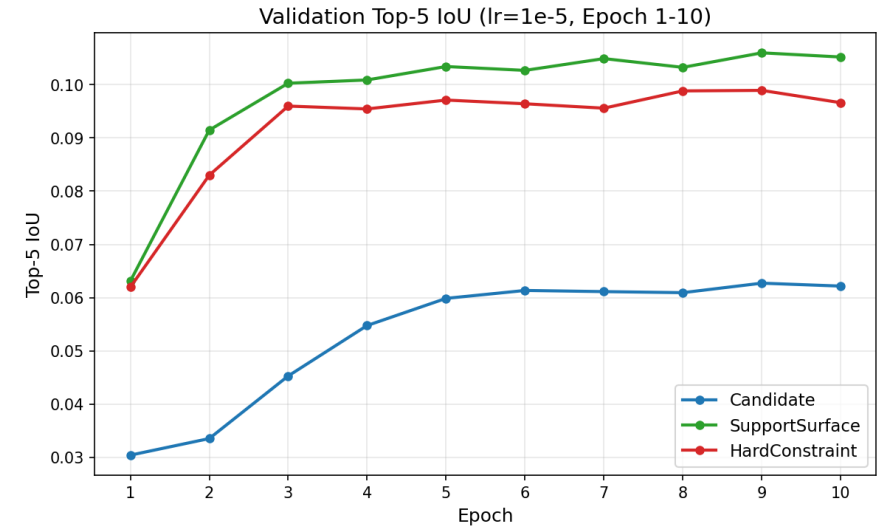
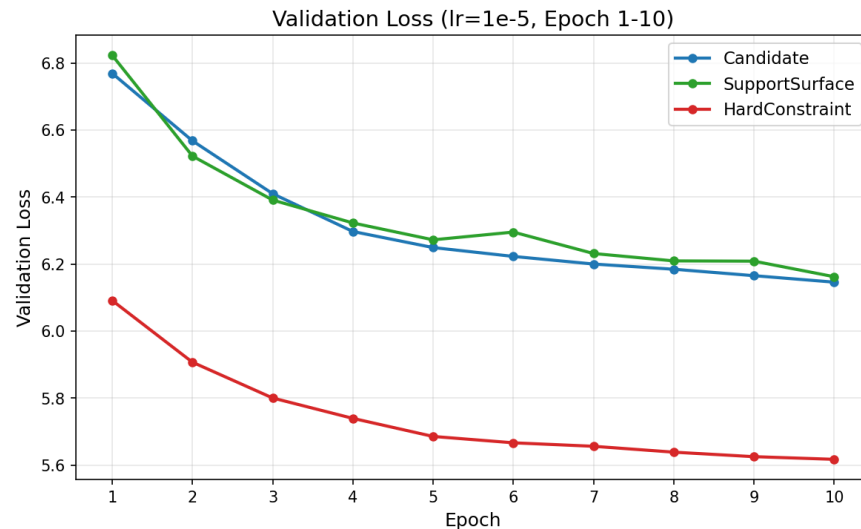
Semantic-guided generation

+ Hard
Constraint

HardConstraint Model

Validity mask filters
implausible candidates before
scoring

Strict physical grounding



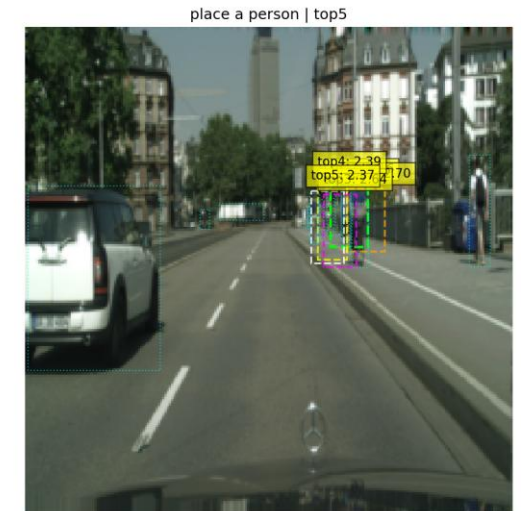
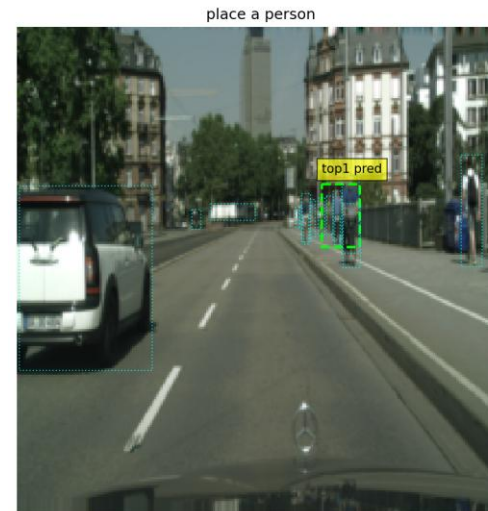
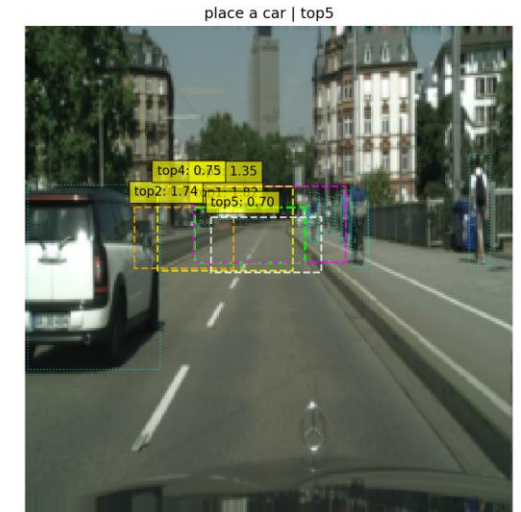
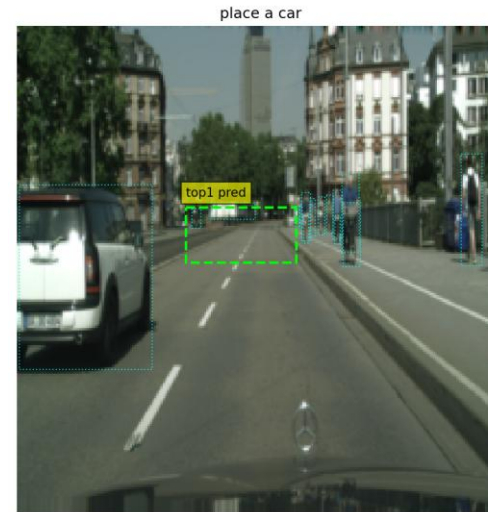
Quantitative Results

Same image, different prompts:

- “place a car” → road
- “place a person” → sidewalk

Insight:

- Model learns object–scene relationships
- Placement depends on prompt
- Framework is context-aware



Qualitative (Ablation Effect)

SupportSurface model:

Candidates sampled from both roads and sidewalks

Model may place pedestrians on road regions

Results can appear visually unrealistic

HardConstraint model:

Invalid candidates are filtered before scoring

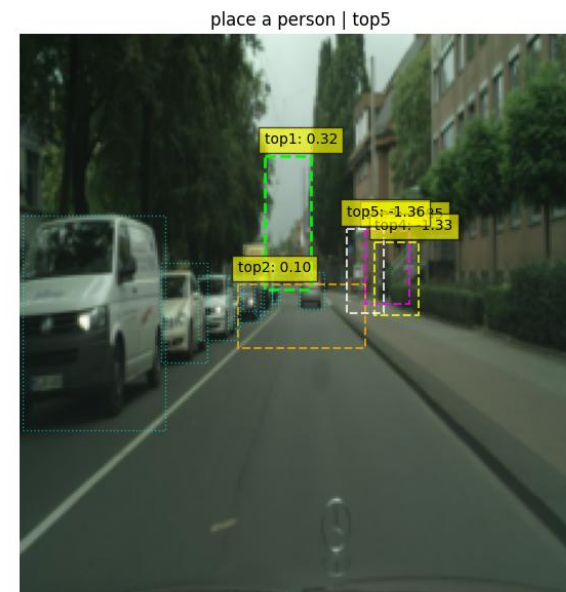
Predictions consistently fall on sidewalk regions

Aligns better with real-world expectations

Key Insight:

Geometric constraints guide the model toward physically plausible placements

Improve semantic realism of predictions



Failure Case Analysis



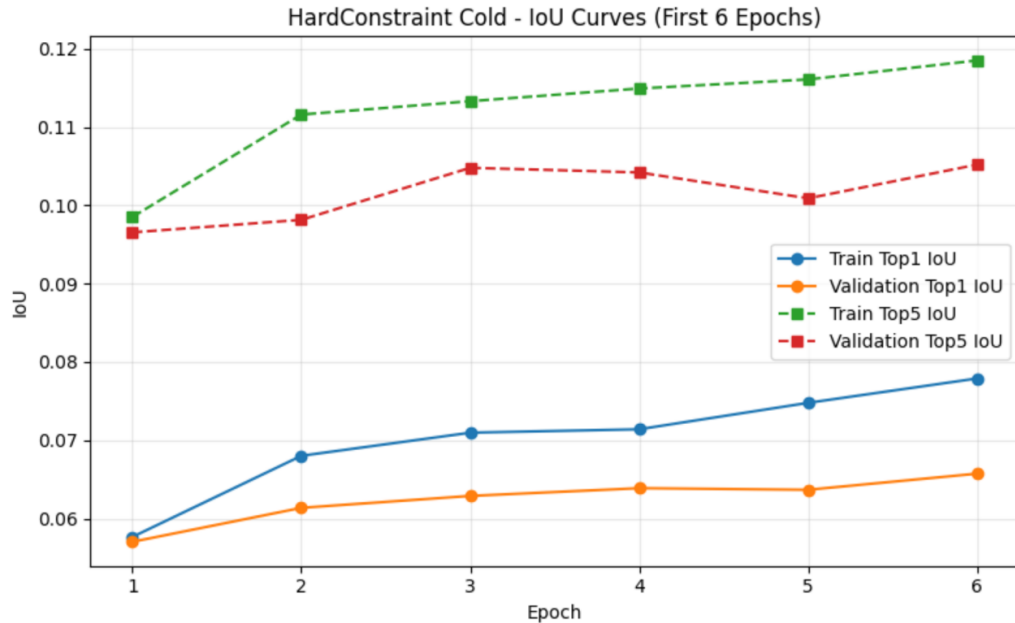
Failure Case: Overlapping Placement

Model predicts a valid road region
But overlaps with an existing vehicle
Leads to unrealistic placement

Cause:

No depth or occupancy reasoning
Only relies on 2D image features

Limitations



Limitations

1. IoU Limitation

- Measures geometric overlap only
- Does not reflect semantic plausibility
- Better placements may get lower IoU

2. Dataset Bias

- Ground truth from object removal
- May introduce artifacts
- Model may learn dataset-specific cues

3. Lack of Depth Awareness

- No 3D understanding
- Cannot reason about occupancy
- Leads to overlapping predictions

Conclusion & Future Work

Conclusion

- Candidate ranking is effective for object placement
- SupportSurface improves semantic consistency
- HardConstraint enforces physically plausible placements
- Produces more realistic results than baseline

Future Work

- Develop metrics beyond IoU (semantic / human-aware)
- Improve dataset construction to reduce bias
- Incorporate depth or 3D reasoning for spatial understanding

Thank you!

Candidate Generation

Generate candidate boxes around the support surface, ~~with a small number of global fallback candidates~~

Hard Constraint

Check plausibility of geometry and class-specific semantic

- Bottom-center location
- Bottom-strip overlap

Why not?

Hard Candidate Generation

Generate candidate boxes based on support surface and class-specific semantic

- Bottom-center location
- Bottom-strip overlap

- Too rigid, may reduce candidate coverage
- Sensitive to noisy support surface
- Fallback candidates improve robustness

Contribution & Novelty

Member	Main Contribution
Zixi Chai	Developed the candidate preparation pipeline
Qihang Feng	Developed the ranking neural network
Hongtao Ren	Designed the evaluation method and analysis

What we borrowed from BOOTPLACE (Hang Zhou et al., 2025)?

- Training data construction idea
- Supervision from object locations
- Placement as candidate-based location

How our method differs from BOOTPLACE?

- Support-surface-guided candidate generation
- Hard constraints for plausibility control
- Different ranking neural network